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PRESSING IRONS

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FIG. 1

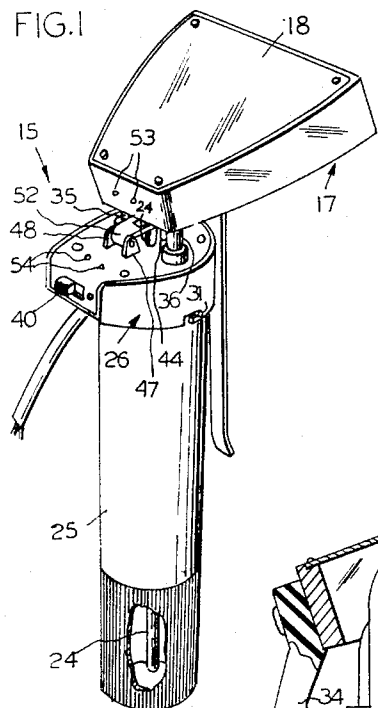


FIG. 2

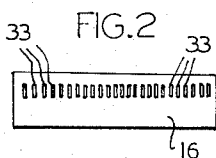


FIG. 3

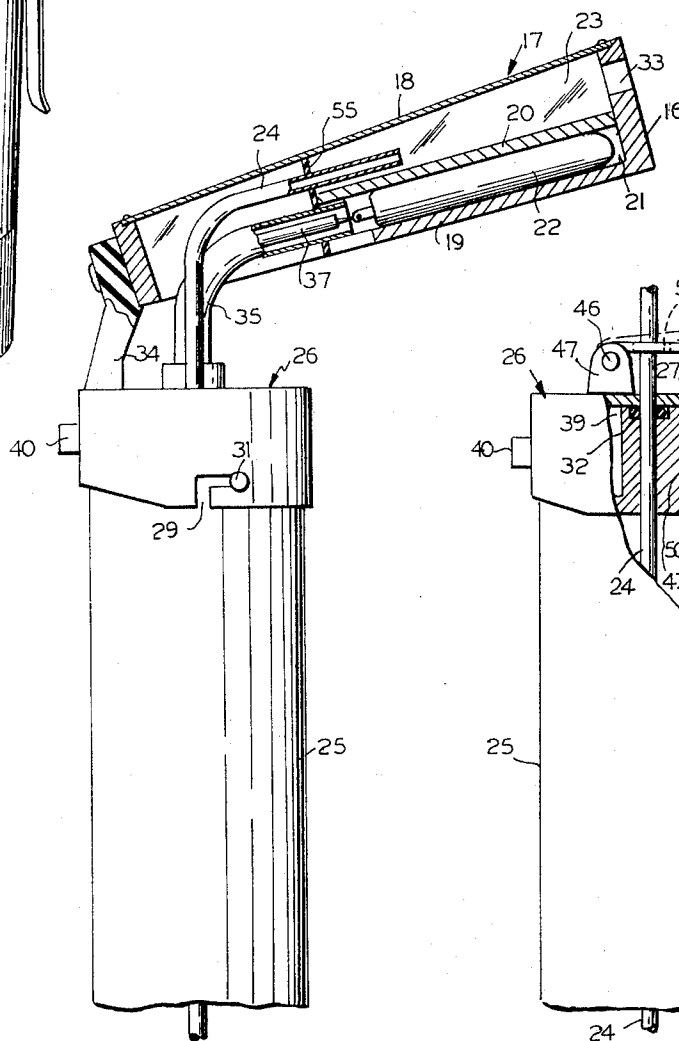
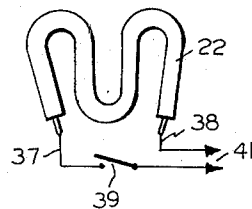


FIG. 4

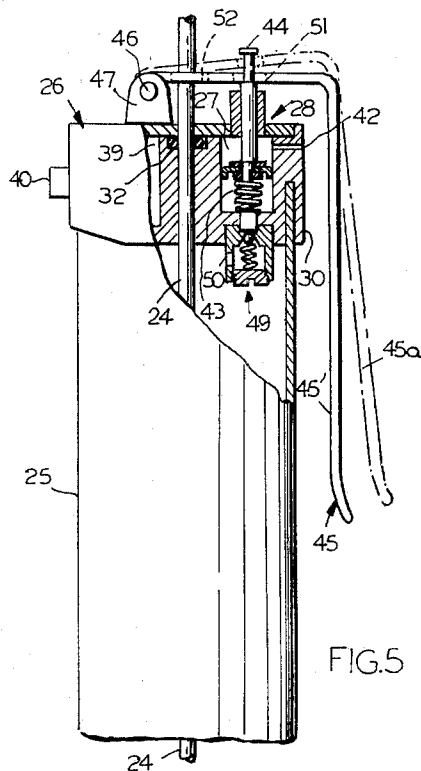


FIG. 5

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PRESSING IRONS

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ABSTRACT OF THE DISCLOSURE

The perforated front wall of a casing is the pressing surface of this device. Said casing overhangs in a forwardly direction and is carried by a releasable head serving as the cover for an upright water tank which is the handle of the device. An air pump on said cover, for pressurizing the water, is operated by grasp and release movements of the handle-holding hand, whereby water is forced upwardly through a tube within the tank, and discharged as a spray into said upper casing wherein an electrically-heated flash boiler vaporizes it, whereupon the steam generated is emitted from said perforations in the forward ironing surface. Said cover member also carries a switch to control the circuit of the heating element.

The present invention relates to, and its principal object is to provide a novel and improved hand-held, electrically heated, steam emitting pressing iron, adapted especially for touch up work on a hanging garment. For example, this iron is to flatten wrinkles or to accentuate a crease in garments taken out of luggage or out of a closet or off a display rack. It is also useful to freshen trouser creases and for last minute work to be done on gowns being fitted.

Another object thereof is to provide an electric steam iron of novel and improved construction of the character set forth, which is compact, simple in structure, reasonable in cost to manufacture, easy to use and efficient in carrying out the functions for which it is designed.

Other objects and advantages will become apparent as this disclosure proceeds.

For one practice of this invention, the pressing surface is offered by the perforated front wall of a comparatively thin casing which overhangs in a forwardly direction and is carried by the releasable head serving as the cover for an upright water tank which is the handle of the device. An air pump on said cover, for pressurizing the water, is operated by grasp and release movements of the handle-holding hand, whereby water is forced upwardly through a tube within the tank, and discharged as a spray into said upper casing wherein an electrically heated flash boiler vaporizes it whereupon the steam generated is emitted from the perforations in the forward ironing surface. Said cover member also carries a switch to control the circuit of the heating element.

In the accompanying drawing forming part of this specification, similar characters of reference indicate corresponding parts in all the views.

FIG. 1 is a perspective view of a pressing iron embodying the teachings of this invention. Part of the tank is shown broken through to expose the water delivery tube therein, and the support for the upper casing is omitted in order to expose various parts which otherwise would be hidden from view.

FIG. 2 shows the ironing surface.

FIG. 3 is a diagrammatic view of the electric heater

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element and the electrical circuit for its operation and control.

FIG. 4 is an enlarged side view of the device shown in FIG. 1, partly in section. The pump-operating lever and its mount are omitted to attain clarity of illustration.

FIG. 5 is similar to FIG. 4, showing the pump means in section, but omitting various parts in order to attain clarity of illustration. This view as well as FIG. 4, are fragmentary.

The ironing device shown in the drawing as an example of this invention, is designated generally by the numeral 15. The ironing surface is the outer face of the flat front wall 16 of a metal casing denoted generally by the numeral 17. This casing has flat top and bottom walls, denoted respectively as 18 and 19, between which there is a metal partition 20, extending from said front wall 16, for a distance rearward inside the casing, thereby forming a lower space 21 for housing the electric heater element 22, and an upper space 23 in which water forced up the delivery tube 24 from the tank 25, is sprayed. Said tank, serving as the handle for the device 15, is an upright elongated receptacle, having a releasable top cover or head, denoted generally by the numeral 26, having a downward bore 27 which serves as the cylinder of an air pump indicated generally by the numeral 28. The attachment means for said head to the tank, may be by a bayonet lock arrangement comprising the L-shaped slot 29 in the apron 30 of the head 26, engaged by the pin 31 extending laterally therein from the tank wall. The water delivery tube 24, extends from near the bottom of the tank, up through the head 26 through a leak-avoiding rubber O-ring 32, housed in a suitable depression, and thence into the space 23, where spray therefrom is confined and is vaporized by the hot casing and particularly by the hot plate 20, in the manner as occurs in a flash boiler. The steam thus generated is discharged through the perforations 33 in the casing's front wall 16; said perforations being communicative with the space 23. Said casing 17 is supported on a heat-insulative member 34 which is fixed atop the cover 26, and extends therefrom forwardly beyond the tank 25. The casing 17 is preferably tilted forwardly upwardly a bit as shown, and into it from ducts 35, 36, enter the conductors 37 and 38 for connection to the terminals of the electric heater element 22. In a suitable cavity in the cover 26, a switch 39 is housed, with its operating element 40 accessible for movement to "on" and "off" positions, to control the circuit containing said heater element and switch in series across a suitable plug not shown, for connection to a power outlet whose terminals are indicated at 41.

The numeral 42 indicates an air inlet vent for the pump, whose piston is biased upwardly by the compression coil spring 43, and whose piston rod 44 is engaged by the operating lever 45 which is pivoted at 46 on bearings 47, 48 carried on the cover 26. Said lever has an arm 45' extending downwardly along the tank 25, adapted to be straddled by the fingers of the hand holding the tank in its palm, so by continued grip and release action of the hand, the pump is operated when it is desired that the pressing done by the ironing surface 16, shall be accompanied by steam ejection through the openings 33. The numeral 49 indicates a check valve, which opens upon downward movement of the pump's piston, allowing air to be forced into the tank through the discharge opening 50, and said check valve automatically closes upon upward movement of said piston and when the piston is at

rest. Suitable openings in the horizontal arm of the lever, as indicated at 51 and 52, afford clearance for the piston rod 44 and the water delivery tube 24 which pass through them respectively. Holes for screws to be associated with the support member 34, are indicated at 53 and 54.

The support 34 may be of Bakelite or a ceramic, the wall 18 of sheet steel, the rest of the casing 17 of cast aluminum and likewise the head or cover member 26, and the tank 25 may be of aluminum or molded plastic. The hot plate 20 may be integral with the ironing wall 16 whose outer surface is preferably smooth and polished. It is also advisable that the bottom wall 19 of the casing shall have an extent of about the hot plate 20, to leave the casing open from the bottom in its rear bottom half in order that the heat shall least effect the tube 24 and the ducts 35, 36 which are supported in suitable holes in the transverse insulative partition 55. This partition closes the chamber 23.

For practical dimensions, the heating surface 16 may measure one inch in height and four inches in length, and the height of the tank may be about eight inches.

In FIG. 5, the pump-operating lever is in position when the part 45' has been pressed towards the tank 25, thus the piston of the pump is in downward position and the spring 43 is tensed in compression. In normal rest position, the operating lever is in a forward position as shown at 45a, and the piston is in raised position.

To use the device, the garment to be worked on is hung up on a hanger high up on a nail extending from the front surface of a closet door, or depending from any other suitable raised support. Below the region to be pressed, the garment is pulled taut. The iron of course has been made hot by closing the switch 39 in a powered circuit, and the tank 25 has been nearly filled with water. Using the tank as a handle, the device 15 is held in hand and the ironing surface 16 is applied to the garment while the user manipulates his fingers for a grasp and release action, working the lever 45 to operate the pump 49. This will pressurize the water in the tank and send a stream up through the delivery tube 24, to be sprayed into the chamber 23 where it is converted into steam which is ejected through the openings 33 onto the garment being pressed. If no steam is desired, the lever 45 is not worked. The underside of the casing wall 19, can be used for ironing on the shoulder of the garment. The bore through the tube 24 is small and its discharge end may terminate in a spraying nozzle not shown, or be flattened to act as a spray nozzle.

A special use for this ironing device 15, is for touching up garments being worn and fitted at the last minute for bridal occasions, in dressing room.

If desired, a small wood block held in hand may be used as an ironing board.

This invention is capable of numerous forms and various applications without departing from the essential features herein disclosed. It is therefore intended and desired that the embodiment shown herein shall be deemed merely illustrative and not restrictive and that the patent shall cover all patentable novelty herein set forth; reference being had to the following claims rather than to the specific description and showing herein to indicate the scope of this invention.

I claim:

1. In an ironing device of the character described, a casing having a metal front wall whose outer surface serves as the ironing surface, an electric heater element housed in the casing, a closed water tank behind said front wall and carrying said casing, a water-delivery tube extending from within the lower part of said tank, then outwardly thereof in water-tight relation therewith and terminating within the casing whereinto it is adapted to discharge, an air pump carried by said device, connected to discharge into the upper part of the tank; said pump having an operating member, where, upon movement thereof, water from the tank will be forced into the

casing through said delivery tube; the heat generated by the heater element when actuated, being for heating the casing and for vaporizing the water entering the casing; said front wall having perforations through which the steam made will pass out of the casing; said water tank forming a handle rearward of said front wall, for holding the device in one hand; the operating member of the pump being accessible and adapted for manipulation by the hand holding the handle, and a metal partition wall within the casing, extending rearwardly from said front wall, forming spaces in the casing, one above and one below said partition wall; the heater element being within one of said spaces, and the delivery tube discharging into the other of said spaces; the perforations through said front wall being communicative with the space receiving the water discharge.

2. An ironing device as defined in claim 1, wherein the heater element is in the lower space.

3. In an ironing device of the character described, a casing having a metal front wall whose outer surface serves as the ironing surface, an electric heater element housed in the casing, a closed water tank behind said front wall and carrying said casing, a water-delivery tube extending from within the lower part of the tank, then outwardly thereof in water-tight relation therewith and terminating within the casing whereinto it is adapted to discharge, and an air pump carried by said device, connected to discharge into the upper part of the tank; said pump having an operating member, where, upon movement thereof, water from the tank will be forced into the casing through said delivery tube; the heat generated by the heater element when actuated, being for heating the casing and for vaporizing the water entering the casing; said front wall having perforations through which the steam made will pass out of the casing; said water tank forming a handle rearward of said front wall, for holding the device in one hand; the operating member of the pump being accessible and adapted for manipulation by the hand holding the handle; said casing having a flat bottom wall of metal, whose bottom surface is adapted to serve as an ironing surface.

4. In an ironing device of the character described, a casing having a metal front wall whose outer surface serves as the ironing surface, an electric heater element housed in the casing, a closed water tank behind said front wall and carrying said casing, a water-delivery tube extending from within the lower part of the tank, then outwardly thereof in water-tight relation therewith and terminating within the casing whereinto it is adapted to discharge, and an air pump carried by said device, connected to discharge into the upper part of the tank; said pump having an operating member, where, upon movement thereof, water from the tank will be forced into the casing through said delivery tube; the heat generated by the heater element when actuated, being for heating the casing and for vaporizing the water entering the casing; said front wall having perforations through which the steam made will pass out of the casing; said water tank forming a handle rearward of said front wall, for holding the device in one hand; the operating member of the pump being accessible and adapted for manipulation by the hand holding the handle; said tank being an open-top receptacle closed by a cover head releasably mounted thereon and in air-tight relation therewith; the casing and pump being carried by said head, and the water delivery tube being through said head.

5. In an ironing device of the character described, a casing having a metal front wall whose outer surface serves as the ironing surface, an electric heater element housed in the casing, a closed water tank behind said front wall and carrying said casing, a water-delivery tube extending from within the lower part of the tank, then outwardly thereof in water-tight relation therewith and terminating within the casing whereinto it is adapted to discharge, and an air pump carried by said device,

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connected to discharge into the upper part of the tank; said pump having an operating member, where, upon movement thereof, water from the tank will be forced into the casing through said delivery tube; the heat generated by the heater element when actuated, being for heating the casing and for vaporizing the water entering the casing; said front wall having perforations through which the steam made will pass out of the casing; said water tank forming a handle rearward of said front wall, for holding the device in one hand; the operating member of the pump being accessible and adapted for manipulation by the hand holding the handle; said handle extend-

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ing downward with respect to the casing and the casing being flat and having a slight slant forwardly upwardly.

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